W-D-1114B
June 14, 1995
SUPERSEDING
W-D-1114A
September 17, 1970

FEDERAL SPECIFICATION

DRILLS, ELECTRIC, PORTABLE, DOUBLE-INSULATED

(DIELECTRIC HOUSING)

This specification is approved by the Commissioner, Federal Supply Service, General Services Administration for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers hand guided, double-insulated, dielectric housing, portable electric drills.

1.1.1 Federal specification coverage. Federal specifications do not cover all varieties of the commodity indicated by the title of the specification, or which are commercially available, but are intended to cover only those generally used by the Federal Government.

1.2 Classification. Double-insulated electric drills covered by this specification shall be of the general service type furnished in the following classes, sizes, and styles as specified (see 6.2):

Class A - Heavy Duty

Size 1/4 inch capacity, single speed

- Style 2 - Pistol grip with auxiliary side handle or side handle with auxiliary side handle (see Figure 2a or 2b)
- Style 3 - End handle, closed grip (see Figure 3)
- Style 5 - End handle, spade or closed grip with side handle and auxiliary side handle (see Figure 5)

Size 3/8 inch capacity, single speed

- Style 2 - Pistol grip with auxiliary side handle or side handle with auxiliary side handle (see Figure 2a or 2b)
- Style 3 - End handle, closed grip (see Figure 3)
- Style 4 - End handle, closed grip and auxiliary side handle (see Figure 4)
- Style 5 - End handle, spade or closed grip with side handle and auxiliary side handle (see Figure 5)

Size 1/2 inch capacity

- Style 4 - End handle, closed grip and auxiliary side handle (see Figure 4)
- Style 5 - End handle, spade or closed grip with side handle and auxiliary side handle (see Figure 5)

FSC 5130

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Class B - Standard Duty

Size 1/4 inch capacity, single speed

Style 1 - Pistol grip (see Figure 1)
Style 2 - Pistol grip with auxiliary side handle or side handle with auxiliary side handle (see Figure 2a or 2b)

Size 3/8 inch capacity, single speed

Style 2 - Pistol grip with auxiliary side handle or side handle with auxiliary side handle (Figure 2a or 2b)

Size 3/8 inch capacity, two speed

Style 2 - Pistol grip with auxiliary side handle or side handle with auxiliary side handle (see Figure 2a or 2b)

Style 5 - End handle, spade or closed grip with side handle and auxiliary side handle (see Figure 5)

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specification:

PPP-P-40 - Packaging and Packing of Hand Tools; Tools and Tool Accessories for Power Driven, Metal, and Woodworking Machinery

Federal Standard:

FED-STD-H28 - Screw-Thread Standards for Federal Services
FED-STD-123 - Marking for Domestic Shipment (Civilian Agencies)

Commercial Item Description:

A-A-51140 - Chucks, Drill, Key Type, Three-Jaw Chuck, Geared

(Activities outside the Federal Government may obtain copies of Federal specifications, standards, and Commercial Item Descriptions as outlined under General Information in the Index of Federal Specifications, Standards and Commercial Item Descriptions. The Index, which includes cumulative bimonthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

(Single copies of this specification, and other Federal specifications and Commercial Item Descriptions required by activities outside the Federal Government for bidding purposes are available without charge from General Services Administration Business Service Centers in Boston, MA; New York, NY; Philadelphia, PA; Washington, DC; Atlanta, GA; Chicago, IL; Kansas City, MO; Fort Worth, TX; Houston, TX; Denver, CO; San Francisco, CA; Los Angeles, CA; and Seattle, WA.)

(Federal Government activities may obtain copies of Federal standardization documents and the Index of Federal Specifications, Standards and Commercial Item Descriptions from established distribution points in their agencies.)

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Military Specification:

MIL-V-173	-	Varnish, Moisture- and Fungus-Resistant (For the Treatment of Communications, Electronic and Associated Electrical Equipment)

Military Standards:

MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-461	-	Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference
MIL-STD-462	-	Electromagnetic Interference Characteristics, Measurement of

(Copies of military specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Non-Government publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American National Standards Institute (ANSI):

ANSI/AFBMA B1	-	Terminology or Nomenclature for Various Types of Bearings and Parts Definitions and Illustrations
ANSI/AFBMA B10	-	Balls: Terminology, Definitions and Tolerances for Metal Balls
ANSI/AFBMA B18.2	-	Needle Roller Bearings, Radial, Inch Design Identification Code, Boundary, Tolerances, Fitting, and Mounting Practices

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

American National Standards Institute (ANSI)/American Society of Quality Control (ASQC):

ANSI/ASQC Z1.4	-	Sampling Procedures and Tables for Inspection by Attributes

(Application for copies should be addressed to the American Society for Quality Control, P. O. Box 3005, 611 E. Wisconsin Avenue, Milwaukee, WI 53201-4606.)

Underwriters' Laboratories, Inc. (UL):

UL 45	-	Electric Tools

(Application for copies should be addressed to the Underwriters' Laboratories, Inc., 333 Pfingsten Road, Northbrook, IL 60062-2096.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Illustrations. The illustrations herein are descriptive and not restrictive and are not intended to preclude the purchase of drills otherwise conforming to this specification.

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3.2 First article inspection. When specified (see 6.2), the contractor shall furnish a completely fabricated portable electric drill for first article inspection in accordance with section 4. The sample shall be a preproduction item or a production item which meets the requirements of this specification. In either case, the sample shall be identical with the proposed items and shall be made or shall have been made by the methods that are to be used in the production of items to fulfill the terms of the contract. Approval of the first article shall not relieve the contractor of the responsibility to furnish equipment in accordance with the requirement of this specification.

3.3 Material. All materials used in the fabrication of double-insulated electric drills shall be new and of the quality necessary to produce drills to meet the requirements described herein. Electrical materials and components shall conform to the Underwriters' Laboratories (UL), Inc., Standard Nos. 45.

3.4 Design. The drills furnished shall be of the manufacturer's current design and shall include industry's latest developments in the double-insulated, dielectric housing tool field. The drills shall be of the standard commercial portable electric drill design. All parts subject to wear shall be easily accessible for adjustment, replacement, and repair.

3.4.1 Conformance to Underwriters' Laboratories (UL), Inc. standards. The drill shall comply with the requirements of the specified Underwriters' Laboratories (UL) standards. The contractor shall submit proof to the procuring activity that the drill complies to the requirements of the applicable UL standards. Compliance shall be indicated by the permanent attachment of an appropriate label, seal, or stamp that contains the UL listing mark to the product, or by a certificate furnished therewith. In lieu of the aforementioned, the contractor shall submit independent proof, satisfactory to the Government, that the drill complies to the requirements of the applicable UL standards including the UL method of test. Contractors compliance with the UL requirements does not absolve the contractor from complete compliance with the requirements of this specification. In the event of difference between any of the requirements of this specification and those of UL, the requirements of this specification shall govern. When electromagnetic interference (EMI) suppression is required, only the basic drill, prior to modification for EMI suppression, shall comply to the requirements of the applicable UL standard.

3.4.2 Lubrication. The bearings, gears, and moving parts shall be assembled with lubricating medium. If lubricant replenishment is required, external lubricant fittings or other suitable means shall be provided. Sealed-for-life bearings and permanently lubricated gear boxes shall not be required to have external fittings. The gears shall operate in a lubricant which shall be prevented from excessive leakage from the gear casing or entering the motor enclosure. Parts requiring protection from dust shall be enclosed.

3.4.3 Interchangeability. All parts from the same manufacturer shall be manufactured to such standards as will permit replacement or adjustment without modification of the parts or drills.

3.5 Construction. Construction shall be free of any characteristics or defects that will prevent the drills from passing any of the examinations and tests specified in section 4. The drills shall be complete so that when connected to the specified source of power, they can be used for any operation for which they were designed.

3.5.1 Fastening devices. All screws, pins, bolts, and similar parts shall be installed with a means for preventing loss of tightness. All such parts when subject to removal or adjustment shall not be swaged, peened, staked, or permanently deformed. All threaded parts shall be in the inch system and shall conform to FED-STD-H28.

3.6 Sizes and capacities. The drills shall comply with the sizes and capacities specified in Table I for the class and size specified (see 1.2 and 6.2). When specified (see 6.2), the drills shall be capable of reversible operation.

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Table I. Drill sizes and capacities

Size of drill and nominal capacity (inch)	Chuck capacity (min) range (inch)	Weight without cable but with handles (max) (pounds)	Revolutions per minute of spindle		Rated-load current at 120 volts AC (amperes) (min)	Torque at spindle shaft (min.)	
			Rated-load rpm range	No load 1/ rpm range		Rated-load rpm (foot-pounds)	At stall (foot-pounds)
Class A, heavy duty, single speed							
1/4	0-1/4	8.5	1000-1300		2.5	1.0	5.0
3/8	0-3/8	15.0	465-575		5.0	3.0	15.0
1/2	5/64- 1/2	21.0	200-400		5.0	5.5	27.5
Class B, standard duty, single speed							
1/4	0-1/4	6.5	1000-1800		1.8	0.5	3.0
3/8	0-3/8	10.5	450-660		2.5	2.0	10.0
Class B, standard duty, two speed							
			(min) Low	(min) High		Low	High
3/8	0-3/8	10.5	550	1600	2.5	2.0	0.75

						Low	High	Low	High
3/8	0-3/8	10.5	550	1600	2.5	2.0	0.75	10	4.5

1/ Spindle speed - No-load spindle rpm shall not exceed 1.82 times rated-load rpm.

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3.7 Styles. Styles are illustrated by the following figures:

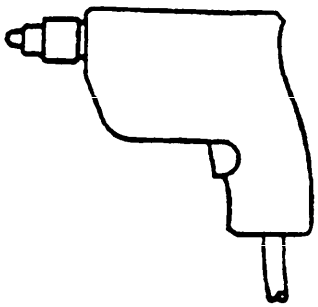


Figure 1 - Style 1, Pistol grip.

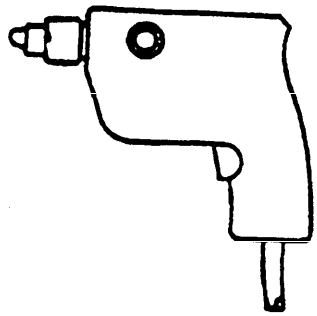


Figure 2a - Style 2, Pistol grip with auxiliary handle.

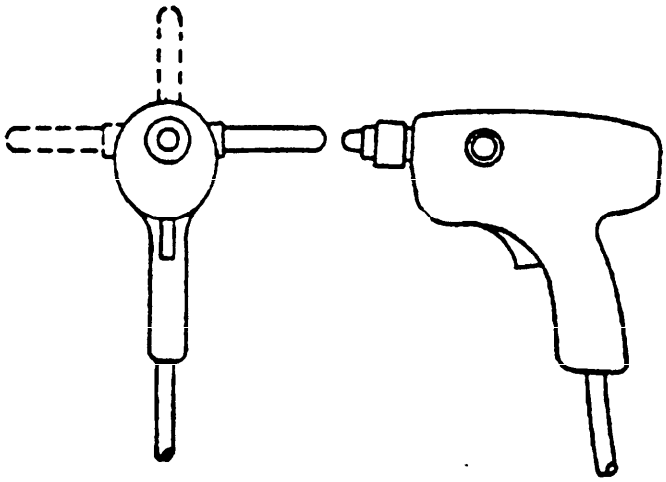


Figure 2b - Style 2, side handle with auxiliary side handle.

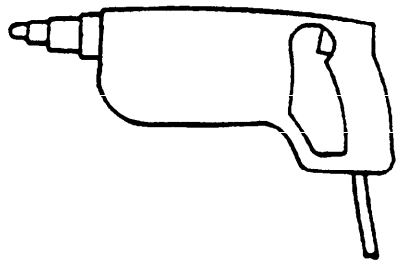


Figure 3 - Style 3, end handle, closed grip.

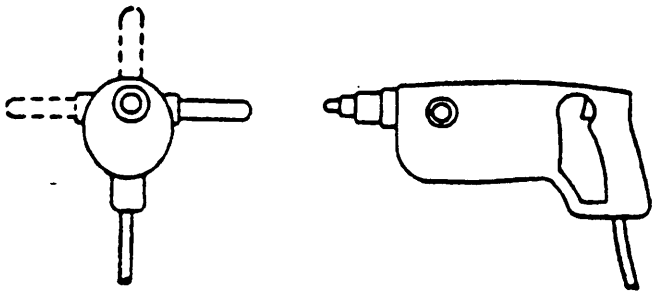


Figure 4 - Style 4, end handle, closed grip and auxiliary handle.

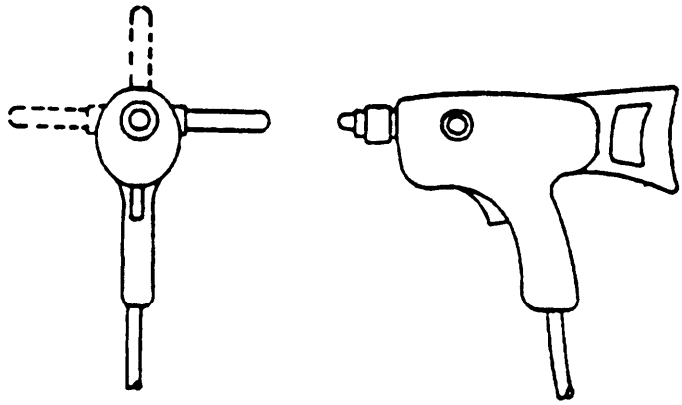


Figure 5 - Style 5, end handle, spade, or or closed grip with side handle and auxiliary side handle.

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3.7.1 Auxiliary side handle. Auxiliary side handles shall be removable, when required, and shall be designed for attachment in at least two of the three positions shown in the applicable figures. Front view, figure 2b is applicable to figure 2a.

3.8 Components. The drills shall be of the materials specified and shall consist essentially of housings, including integral or separately attached handle, chuck, motor, switch, wiring, cord, and plug cap.

3.8.1 Housing. The housing shall be of dielectric material that will totally insulate the operator from the current carrying parts of the tool and from the spindle if made live by an external source or a broken or chafed conductor inside the drill. Drill housing shall be of a uniform thickness (± 1/32 inch of manufacturers design thickness). A metallic skeleton or framework may be incorporated to provide the strength, alignment, and a means of radiation suppression where deemed necessary for fulfilling the requirements and tests specified herein. The housing shall not have any exposed metallic bolts, nuts, or screws, or any appurtenances that are not completely insulated from current carrying and internal metal parts by at least 0.080 inches of insulation, by a creepage or clearance distance of at least 3/16 inch or an equal combination of the two. All external metal parts shall be completely insulated from the spindle by at least 0.031 inch of insulation.

3.8.1.1 Molded parts of housing. Flash shall be removed from molded parts. Flash edges shall be smooth. To prevent moisture absorption into the volume of parts made of thermosetting or thermoplastic material, buffed or cut areas may be given two coats of varnish conforming to MIL-V-173.

3.8.2 Motor. The motor shall be of a universal series wound or compensated series wound designed to operate on either 120 volt (± 10 percent) direct current (DC), or 120 volt (± 6 percent), single phase 25 to 60 Hz alternating current (AC). The motors shall conform to the requirement of UL 45.

3.8.2.1 Rotor. The rotor (armature) shall be balanced dynamically in accordance with commercial practice to ensure freedom from destructive vibration.

3.8.2.2 Motor shaft insulation. Secondary or protecting insulation shall be provided between the armature shaft and the lamination core and commutator unless otherwise insulated between gears.

3.8.2.3 Commutator. The commutator segments shall be of hard drawn, hard rolled, or drop forged copper. The commutator shall have sheet mica separators between segments or shall be of the molded mica construction using material that will be immune to deformation and dielectric degradation when subjected to maximum rotational forces operating temperatures encountered in service.

3.8.3 Switch. The drills shall be provided with a conveniently located switch of the quick break design for turning the motor "OFF" and "ON". The switch shall be encased in high dielectric insulating material and shall comply with UL 45 requirements for switches used in double-insulated tools. The inclusion of a locking device or similar means for holding the switch in the "ON" position shall be at the option of the contractor. The switch shall have a current rating of not less than the rated load of the drill and shall be capable of interrupting stalled motor current at rated voltage for 50 consecutive contacts at 10 second intervals without sticking or burning of contacts. The switch location shall be as shown on the applicable figure for the particular style.

3.8.4 Wiring, cord, and plug cap. All wiring and flexible cord shall be in accordance with UL 45. The cord shall have an attachment plug cap and unless otherwise specified (see 6.2), shall have a minimum free length of 8 feet measured from point of entry into the drill.

3.9 Fungus proofing when specified (see 6.2). The electric circuitry, including all components and connections, shall be protected from the effect of water and fungus growth by an overall treatment with a varnish conforming to MIL-V-713 with the following exceptions:

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(a) Components or circuit elements which are inherently fungus and water resistant, or which are hermetically sealed, need not be treated.
(b) Components or circuit elements whose function will be adversely affected by the varnish coating shall not be treated.

The varnish, when used, shall be applied to give a minimum dry film thickness of 0.001 inch to component or element surface previously cleaned and prepared in accordance with accepted commercial practice.

3.10 Electromagnetic interference suppression. When specified (see 6.2), the drills shall be electromagnetic interference free within the limits specified for portable electric equipment in compliance with the requirements of MIL-STD-461 and MIL-STD-462.

3.11 Bearings. All rotating parts of the drills shall be provided with precision quality commercially available ball and roller bearings conforming to ANSI/AFBMA B1, B1, and B18.2. Integral or separate shielding to prevent leakage of the lubricant and entrance of contaminants shall be provided. Steel inserts may be omitted in the mounting of bearings when sufficient contact area and suitable fit between bearing outer-diameter and housing is provided.

3.11.1 Armature shaft bearings. Armature shaft bearings shall be ball, roller, needle roller or a combination thereof providing hardness of shaft; provisions for thrust loading and sealing or shielding are adequate.

3.11.2 Spindle bearings. Spindle arbor bearings shall be either ball, roller, taper roller, or needle roller. If roller or needle roller bearings are used, axial thrust loads applied to the spindle shaft must be provided for by use of a separate thrust bearing, either ball or roller, or an integral combination type of roller or needle roller thrust bearing.

3.12. Gears and shafts. The gears and shafts for class A drills shall be manufactured from hardened alloy steel. The gears and shafts for class B drills may be prehardened medium carbon steels which have proved commercially satisfactory in service. All gears and shafts shall be supported by at least two bearings and shall be of such dimensions as to prevent deflection. The gears shall have machine cut or roll-formed teeth and be designed to efficiently transmit imposed loads. Gears and shafts, except chuck end of spindle shaft, shall be fully enclosed. The speed change gear system (two-speed drills when specified, see 6.2) shall provide convenient and positive preselection of drill speed during no-load operation.

3.13 Chuck. The drills shall be provided with a three-jaw geared type key operated drill chuck of the size and capacity range listed in table I for the size of drill specified. The chuck shall have an internally threaded mounting hole to fit the threaded spindle of the drill. Each chuck shall be furnished with a key and thong or equal means for attaching the key to the cable. Chucks for class A and B drills shall be in accordance with Commercial Item Description A-A-51140, class 2, style B.

3.14 Identification plate. Each drill shall bear an identification plate in accordance with the applicable requirements of UL 45.

3.15 Handbook of instructions. Unless otherwise specified (see 6.2), each drill shall be furnished with an instruction manual in accordance with the applicable requirements of UL 45.

3.16 Tools and extra parts. Each drill shall be furnished with such maintenance tools and extra parts as normally furnished commercially.

3.17 Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within the tolerances specified and all other requirements of this document are met. If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch-pound units, a request should be made to the specification preparing activity for changes to this document.

3.18 Workmanship. Details of workmanship shall be in accordance with the best commercial standards and practices. Paints, coatings, platings, and finishes shall be smooth, dry, adherent, continuous, and not stained or discolored.

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All fasteners shall be firmly secured and show no evidence of deformation, cross threading, or hazardous burrs. Adhesives and lubricants adequate for their intended purpose shall be properly and neatly applied. Adhesives shall be adequately cured. Wires and cables shall be neatly dressed and adequately cured. Wire and cable insulation shall show no evidence of burns, abrading, or pinch marks. There shall be no interference, binding, or galling of parts. External and bearing surfaces shall be free of tool and gouge marks, nicks, or other surface imperfections. The item shall be clean and free of corrosion and debris (e.g., chips, shavings, slivers) or other foreign material. The item shall be free from manufacturing workmanship defects (e.g., loose, missing, binding or misaligned parts, sharp or rough external edges, corners, or surfaces) and material workmanship defects (e.g., pits, rips, fins, burrs, tears, nodules, cracks, blisters) which may adversely impact the item's serviceability, durability, safety, or appearance.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. In accordance with 4.1 above, the supplier is responsible for insuring that components and materials used were manufactured, tested, and inspected in accordance with the requirements of referenced subsidiary specification and standards to the extent specified, or, if none, in accordance with this specification.

4.2 Classification or inspection. Inspection shall be classified as follows:

(a) First article inspection.
(b) Quality conformance inspection.

4.2.1 First article inspection. First article inspection shall be applied to the first article submitted in accordance with 3.2 Failure of the first article to pass the examination of one or more preproduction tests shall be cause for rejection. Unless otherwise specified (see 6.2), first article inspection shall consist of the examination in 4.4.1 and all tests in 4.5.

4.3 Sampling for quality conformance.

4.3.1 Lot. All drills of the same class, size, and style offered for delivery at one time shall be considered a lot for purposes of quality conformance inspection.

4.3.2 Sample. A random sample of drills shall be selected from each lot in conformance with ANSI/ASQC Z1.4 at inspection level S-4.

4.4 Quality conformance inspection. Quality conformance inspection shall be performed on each sample drill selected in accordance with 4.3.2. Quality conformance inspection shall consist of the following:

(a) Examination 4.4.1
(b) Tests 4.4.2
(c) Inspection of preparation for delivery requirements (see 4.6).

4.4.1 Examination. Each drill selected in accordance with 4.3.2 shall be examined to determine compliance with the requirements of this specification. Examination shall be conducted in accordance with table II, acceptable quality level (AQL) to be as indicated. Nonconforming drills shall not be offered for delivery and if the nonconforming drills in any one sample exceed the acceptance number for that sample, the lot represented shall be rejected.

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Table II. Classification of defects

Examine	Defect	Critical	Major	Minor
Construction and workmanship	AQL 0.10 percent defective.	001		
	Moving parts (hazardous to operator) not adequately guarded with guards which do not interfere with normal operation of drill			
	AQL 2.5 percent defective		101	
	Not manufacturer's model designation as listed on Qualified Products List			
	Class and size of drill not as specified		102	
	Style of handle not as specified		103	
	Motor not type specified (see nameplate)		104	
	Cord and plug not as specified		105	
	Evidence of leakage of lubrication from gear housing		106	
	Chuck not type specified, key and thong missing		107	
	Chuck defective, fails to firmly and centrally hold straight drills of the required sizes		108	
	Evidence of vibration or excessive noise at no-load speed		109	
	Drill incomplete, parts missing, or not securely assembled		110	
	Component parts defective: Bent, cracked, or not free from sharp edges, fins or burrs		111	
	Brush screw caps (if used) not protected from mechanical injury		112	
	Switch defective, loose, sticks, or binds, not rigidly mounted or switch not provided with locking device for holding it in "ON" position, or not covered with dielectric (insulated material) externally		113	
	Pressure lubricating fittings used on motor bearings		114	
	Electromagnetic interference suppression not included as specified		115	
	Fungus proofing not included when specified		116	
	AQL 6.5 percent defective			
	Instruction books and part lists missing or not as specified			201
	Nameplate not as specified			202

4.4.2 Tests for quality conformance. Each sample selected in accordance with 4.3.2 shall be tested for compliance with the requirements of this specification. AQL shall be 1.5 percent defective. Nonconforming drills shall not be offered for delivery and if the number of defective units exceeds the acceptable number for the samples tested, the lot represented shall be rejected. Quality conformance tests shall consist of the tests specified in 4.5.1, 4.5.2, and 4.5.3.

4.5 Tests.

4.5.1 Electromagnetic interference test report. When specified (see 6.2), the manufacturer shall provide certification that the electromagnetic interference suppression network in the drills offered is identical to that of drills previously tested and approved.

4.5.2 UL 45 tests certification. The offerer shall provide certification that the drill offered conforms to the test requirements of UL 45 (see 3.4.1 for compliance to this requirement).

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4.5.3 Speed, amperage, and other tests. Sample drills shall be tested to verify that the drills comply with the requirements specified below and as listed on the nameplate. Variation in values shall not be more than percentages allowed by UL 45. Nonconforming drills shall be rejected:

(a) No-load and rated-load revolutions per minute.
(b) Rated load amps at nameplate voltage.
(c) Rated load and stalled torque at spindle shaft.

4.6 Inspection of preparation for delivery requirements. An inspection shall be made to determine that the preservation, packaging, packing, and marking comply with the requirements in section 5 of this specification. Defects shall be scored in accordance with table III. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

Table III. Classification of preparation for delivery defects

Examine	Defect
Marking (exterior and interior)	Omitted; incorrect; illegible; improper size, location, sequence, or method of application
Materials	Any component missing, damaged, or not as specified
Workmanship	Inadequate application of components such as incomplete closure of container flaps, loose strapping, inadequate stapling Distortion of container
Contents	Number per container is more or less than required Net weight exceeds requirements

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, and packing. Preservation, packaging, and packing shall be in accordance with PPP-P-40. The level of preservation, packaging, and packing shall be A, B, or Commercial, as specified (see 6.2).

5.2 Marking.

5.2.1 Civil agencies. In addition to any special marking specified in the contract or order, each unit package and shipping container shall be marked in accordance with FED-STD-123.

5.2.2 Military agencies. Interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Intended use. The double-insulated electric drills covered by this specification are intended to perform all drilling operations of like size metallic case housing drills. These drills are particularly recommended for use aboard ships, in damp locations, or outside locations because of their design.

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6.2 Ordering data. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Class, size, style, and capacity required (see 1.2 and 3.6).
c. When drills are to be reversible (see 3.6).
d. First Article Inspection when required (see 3.2 and 4.2.1).
e. Cord-length if different (see 3.8.4).
f. Fungus proofing when required (see 3.9).
g. Electromagnetic interference suppression when required (see 3.10 and 4.5.2).
h. Two-speed drill, when required (see 3.12).
i. Level of preservation and packaging and level or packing required (see 5.1).
j. Marking if other than specified.

6.3 Definitions. Definitions are as follows:

6.3.1 Double-insulation. Insulation comprising both functional insulation, and protecting insulation, with the two insulations physically separated and so arranged that they are not simultaneously subjected to the same deteriorating influences (temperatures and contaminants) to the same degree.

6.3.1.1 Functional insulation. Insulation necessary for the proper functioning of the tool and for basic protection against electrical shock.

6.3.1.2 Protecting insulation. Denotes an independent insulation provided, in addition to the functional insulation, in order to ensure protection against electric shock in case of failure of the functional insulation. In addition, enclosures (housing) of insulation material shall form a part or the whole of the supplemental insulation.

6.3.1.3 Reinforced insulation. Denotes an improved functional insulation with such mechanical and electrical qualities that it provides the same degree of protection against electric shock as double-insulation. The enclosures of insulating material may form a part or the whole of the reinforced insulation.

MILITARY INTERESTS:

Custodians:

Army - GL
Navy - SH
Air Force - 84

Review Activities:

Army - EL, GL
Navy - SH
Air Force - 84

User Activities:

Army - CE, ME
Navy - YD

MILITARY COORDINATING ACTIVITY:

Army - GL

Preparing Activity:

GSA - FSS

Project No. 5130-0133

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